# Bending Arcs of Antimuon Boosting Ring

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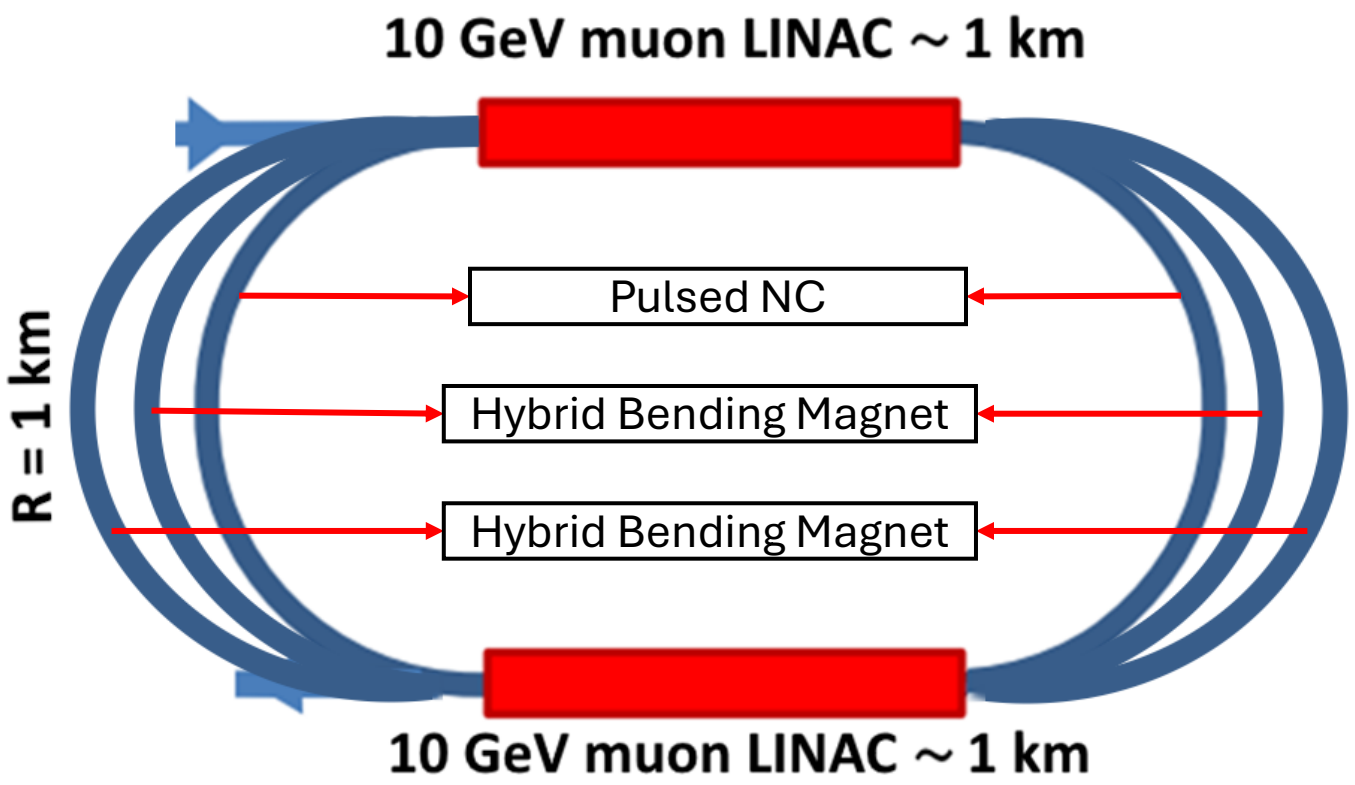
16/12/2025

# Recirculating Linear Accelerator (RLA) + Rapid Cycling Synchrotron (RCS) = Rapid Recirculating Synchrotron (RRS)

**Pulsed NC + Steady SC → Hybrid Bending Magnet**




MuCol (2025). MuCol Milestone Report No. 7: Consolidated Parameters. arXiv preprint arXiv:2510.27437.

| Parameter | Unit | RCS1 | RCS2 | RCS3 | RCS4 |
|---|---|---|---|---|---|
| Hybrid RCS | - | no | yes | yes | yes |
| Repetition rate | Hz | 5 | 5 | 5 | 5 |
| Circumference | m | 5990 | 5990 | 10700 | 35000 |
| Injection energy | GeV | 63 | 314 | 750 | 1500 |
| Extraction energy | GeV | 314 | 750 | 1500 | 5000 |

325 GeV muons x 7 TeV protons

$\sqrt{s} = 3\ TeV$

First stage with only pulsed bending magnets?

Ramp rate is needed over 4000 T/s !!

500 GeV muons can be possible only with Hybrid magnets!!

One arcs each side would be enough?

(SC 2T and NC -1.8T to +1.8T)

# Why MC choose single-bunch option?

We have 40 bunches at beam!!

The above luminosity can be roughly estimated analytically using the following assumptions:

- One bunch of $\mu^+$ is colliding against one bunch of $\mu^-$ (as for the same total number of particles, it is more efficient to have all the particles in one bunch),
- Densities are uncorrelated in the three planes,
- Gaussian distributions in the transverse planes,
- Same parameters for both bunches,
- Round (transverse) beam,
- No crossing angle,
- No transverse offset,
- Ignoring the beam-beam enhancement.

The Muon Collider luminosity formula is typically written as

$$L = \frac{N_0^2 f_0 \gamma}{4\pi\beta^* \epsilon_n} F_{\mathrm{HG}}(\beta^*/\sigma_z) F_{\mathrm{decay}} \quad , \qquad \text{(A.5)}$$

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- It looks like the primary reason for the muon collider to choose for a single-bunch scheme appears to be the optimization of luminosity.
- Luminosity scales linearly with the number of bunches, but is proportional to the square of the number of particles per bunch.
- No other reason for choosing the single-bunch scheme is mentioned in the Milestone Report.
- But still multi-bunches will probably be a problem for our design

# Time Structure of accelerator operation

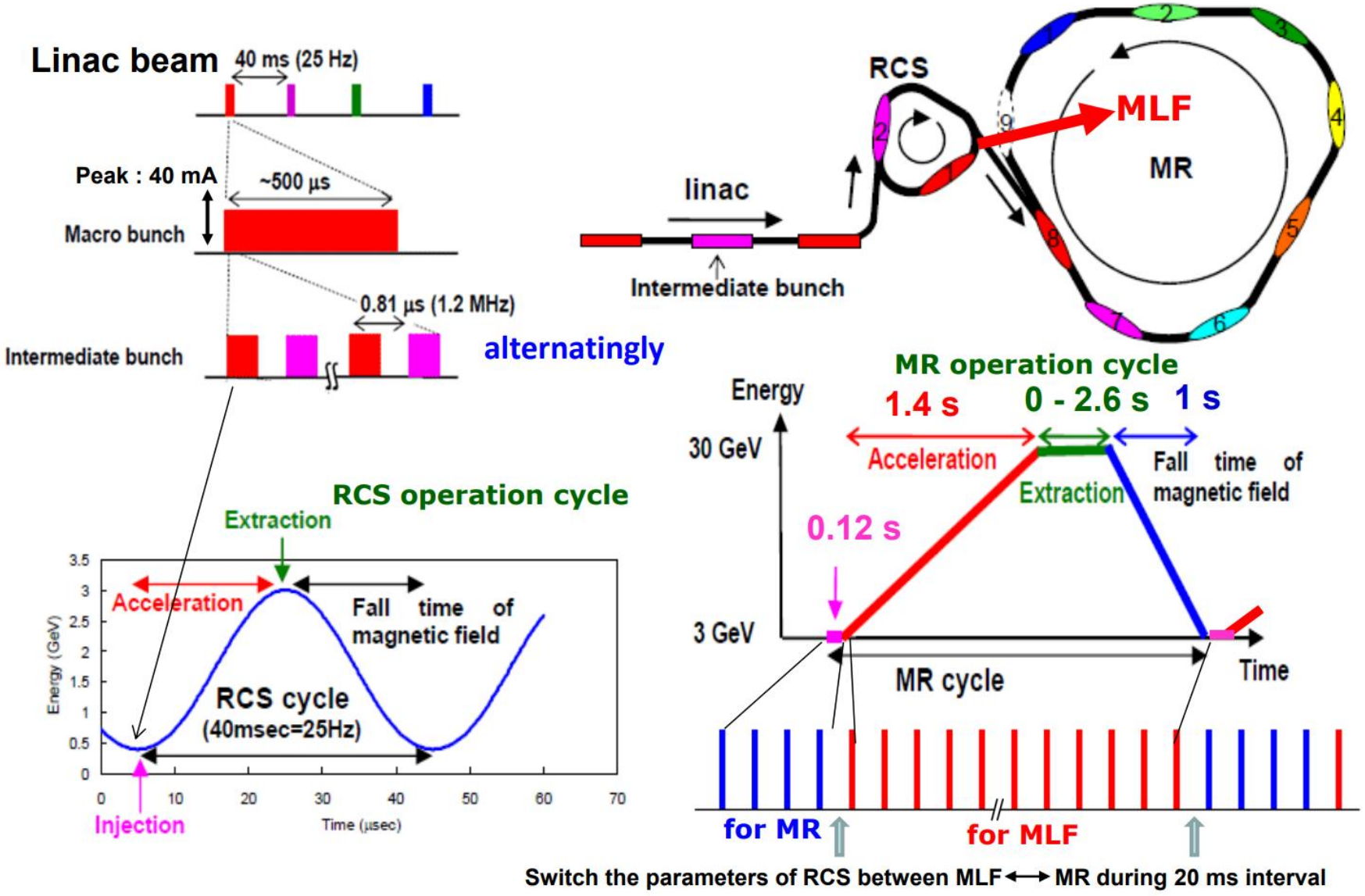
- It looks like RCS merges intermediate bunches into a macrobunch for higher number of protons per bunch.

- Maybe we can do the same for our 40-bunched muons?? Maybe we can merge them??

**Number of bunch ↓**
**Muons per bunch ↑**

μTRISTAN→$10^{10}$  MC→$10^{12}$

$$\frac{\sim 1T}{20ms} = 50\ T/s$$



# Example studies of Pulsed Bending Magnets

Table 1: Specifications for the pulsed bending magnet.
() value is for the second magnet.

| | |
|---|---|
| Max. beam energy | 3 GeV |
| Deflection angle | 114.4 mrad |
| Field strength @ 3 GeV | 1.15 T |
| Magnet core length | 990 (980) mm |
| Magnet core gap & width | 30 x 155 mm |
| Coil turn | 1 |
| Max. PS peak current | 32 kA |
| Peak current @ 3 GeV | 28 kA |
| Pulse width | 200 $\mu$sec |
| Pulse shape | half sine |
| Max. repetition | 25 Hz |
| Effective current | 1.4 kA |
| Stability | $1 \times 10^{-3}$ (p-p) |
| Peak force for coil | 1.6 ton |
| Magnet resistance | 1 m$\Omega$ |
| Magnet inductance | 6.6 $\mu$H |

$$\frac{1.15T}{100\mu s} \approx 10,000\ T/s$$

Tawada, M., Kikuchi, M., Mimashi, T., Nagahashi, S., & Ueda, A. (2009). Development of pulsed bending magnet for simultaneous top-up injection to KEKB and PF ring. Proc. PAC09, Vancouver, BC, Canada, 175.

$$1,000\ T/s$$

**TABLE 3.** Dipole magnet specification

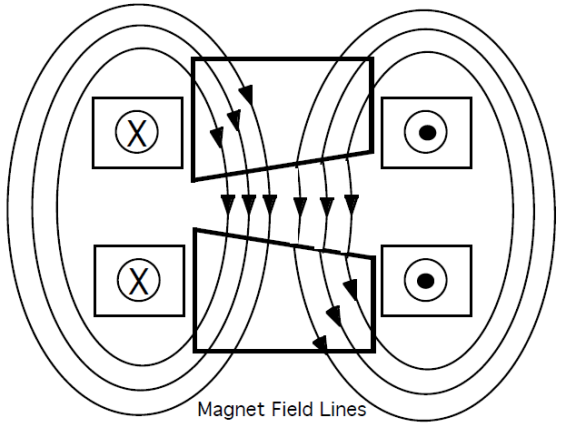
| | | | |
|---|---|---|---|
| Total required dipole length | 2200 m | Beam repetition rate | 15 Hz |
| Gap | 25 mm | Good field region (h x v) | $60 \times 10$ mm$^2$ |
| Design field $B_{design}$ | 1.5 T | Maximum field $B_{max}$ | 1.75 T |
| Field quality at $B_{max}$ | $1 \times 10^{-3}$ | Ramp rate (equivalent frequency) | 1000 Hz |
| Power Loss Yoke (at $B_{design}$) | 2.7 MW | Power Loss Coil (at $B_{design}$) | 0.55 MW |
| Total Power Loss (at $B_{design}$) | 3.25 MW | Stored energy (at $B_{design}$) | 4200 J/m |
| Current per bus bar | 15600 A turns | Number of bus bars | 4 |
| Average peak current density cable | 16 A/mm$^2$ | DC resistance bus bar | $1.77 \times 10^{-5}$ $\Omega$/m |
| Voltage required to drive current | 866 V/m | Inductance | 8.84 $\mu$H |

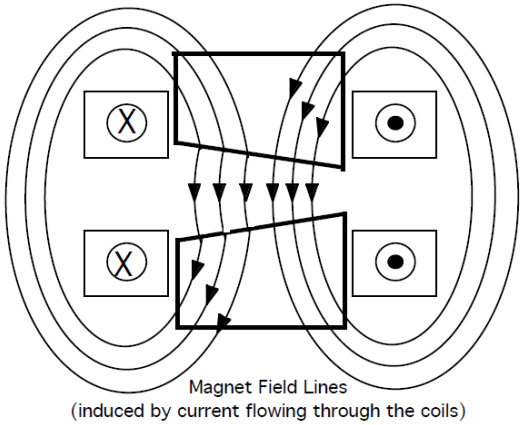
Berg, J. S., & Witte, H. (2016, October). Pulsed synchrotrons for very rapid acceleration. In AIP Conference Proceedings (Vol. 1777, No. 1, p. 100002). AIP Publishing LLC.

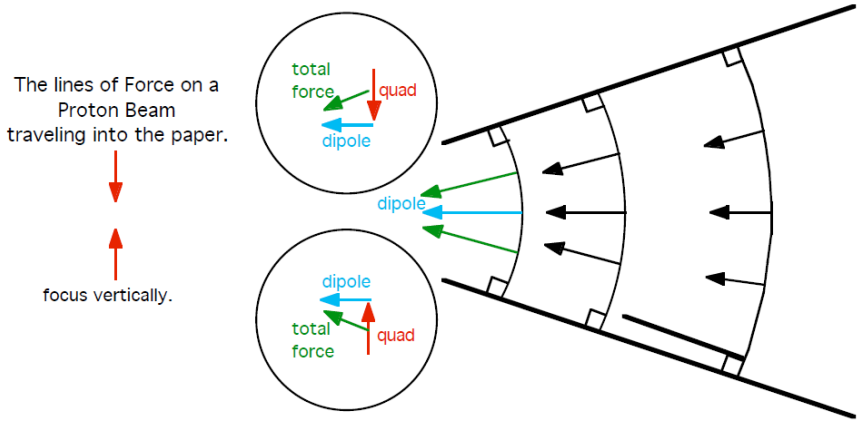
# How alternating gradient bending magnets Works?

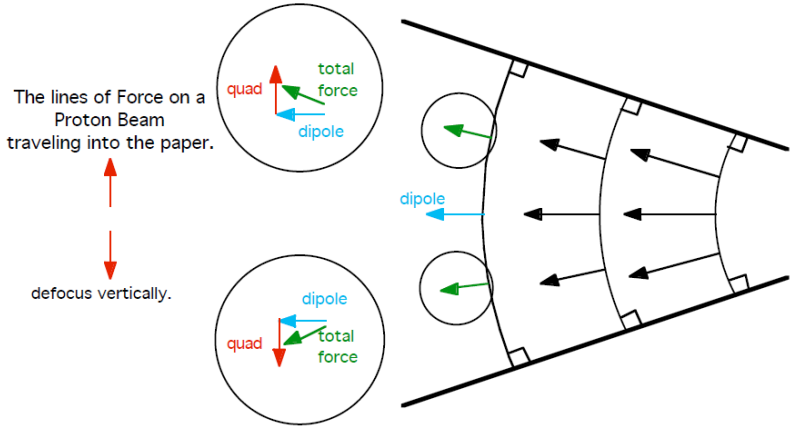
the existing 8 GeV Booster synchrotron [22,23].

The Fermilab Booster [24] is a 474.2 m circumference, alternating-gradient, rapid-cycling synchrotron containing 96 combined-function magnets; see Fig. 1. Together with capacitor banks, these magnets form a resonant network




**Design Report:** https://beamdocs.fnal.gov/AD/DocDB/0010/001022/001/Booster%20V3_0.pdf

# vFFA as alternative for RCS rings of the MC

**Table F.5:** FFA alternative tentative parameters

| Design Parameter | Symbol | Unit | vFFA1 | vFFA4 |
|---|---|---|---|---|
| Orbit radius at centre of F-magnet | r_0 | m | 953 | 5570 |
| F-magnet bending angle | $\theta_F$ | rad. | 0.01033 | 0.00786 |
| Number of cells | N_c | | 790 | 1000 |
| F-magnet half-opening angle | $\beta_F$ | rad. | 0.0015640 | 0.0019700 |
| D-magnet half-opening angle | $\beta_D$ | rad. | 0.0011800 | 0.0011660 |
| F-magnet orbit inclination | $\gamma_F$ | rad. | 0.252 | -0.492 |
| vFFA normalised field index | m | 1/m | 31.94 | 12.13 |
| Comparison Parameter | | | | |
| Circumference [m] | | m | 5990 | 35000 |
| Injection Energy [TeV] | | TeV | 0.06 | 1.5 |
| Extraction Energy [TeV] | | TeV | 0.3 325 GeV ?? | 5 |
| Ramp Rate [T/s] | | T/s | 0 | 0 |
| Vertical Excursion [m] | | m | 0.048 | 0.099 |
| Relative path length difference | | | 0 | 0 |
| Peak Dipole Field On Orbit [T] | | T | 6.93 | 13.59 |
| Peak Dipole Field (Good Field Region) [T] | | T | 12.52 | 29.04 |
| Drift length [m] | | m | 1.18 | 1.03 |
| Tune | | | (0.382, 0.079) | (0.460, 0.057) |

MuCol (2025). MuCol Milestone Report No. 7: Consolidated Parameters. arXiv preprint arXiv:2510.27437.

**Table 33.** 50GeV ERL—quadrupole magnet and RF cavitiy counts along with basic magnet/RF parameters: magnetic-field gradient ($G$), aperture radius ($a$), magnetic length ($L$), frequency ($f$), number of cells per RF cavity (cell), and RF gradient ($G_{RF}$).

| | Quadrupoles | | | | RF cavities | | | |
|---|---|---|---|---|---|---|---|---|
| Section | $N$ | $G$(T m$^{-1}$) | $a$(cm) | $L$(m) | $N$ | $f$(MHz) | Cell | $G_{RF}$(T m$^{-1}$) |
| Linac one | 29 | 7.7 | 3 | 0.25 | 448 | 802 | 5 | 20 |
| Linac two | 29 | 7.7 | 3 | 0.25 | 448 | 802 | 5 | 20 |
| Arc one | 255 | 9.25 | 2.5 | 1 | | | | |
| Arc two | 255 | 17.67 | 2.5 | 1 | | | | |
| Arc three | 255 | 24.25 | 2.5 | 1 | 6 | 1604 | 9 | 30 |
| Arc four | 255 | 27.17 | 2.5 | 1 | 12 | 1604 | 9 | 30 |
| Arc five | 249 | 33.92 | 2.5 | 1 | 18 | 1604 | 9 | 30 |
| Arc six | 249 | 40.75 | 2.5 | 1 | 36 | 1604 | 9 | 30 |
| Total | 1576 | | | | 968 | | | |

**LHeC Design Report**: Agostini, P., et al. "The large hadron–electron collider at the HL-LHC." Journal of Physics G: Nuclear and Particle Physics 48.11 (2021): 110501.

Aperture size of 5 cm looks okey for muLHC with vFFA1 arcs

# Scaling Fixed Field Accelerators: Theory and Modelling of Horizontal- and Vertical-Excursion Accelerators

Max Emil Topp-Mugglestone
Wolfson College

Thesis submitted in fulfilment of the requirements for the degree of Doctor of Philosophy at the University of Oxford

Hilary Term, 2024

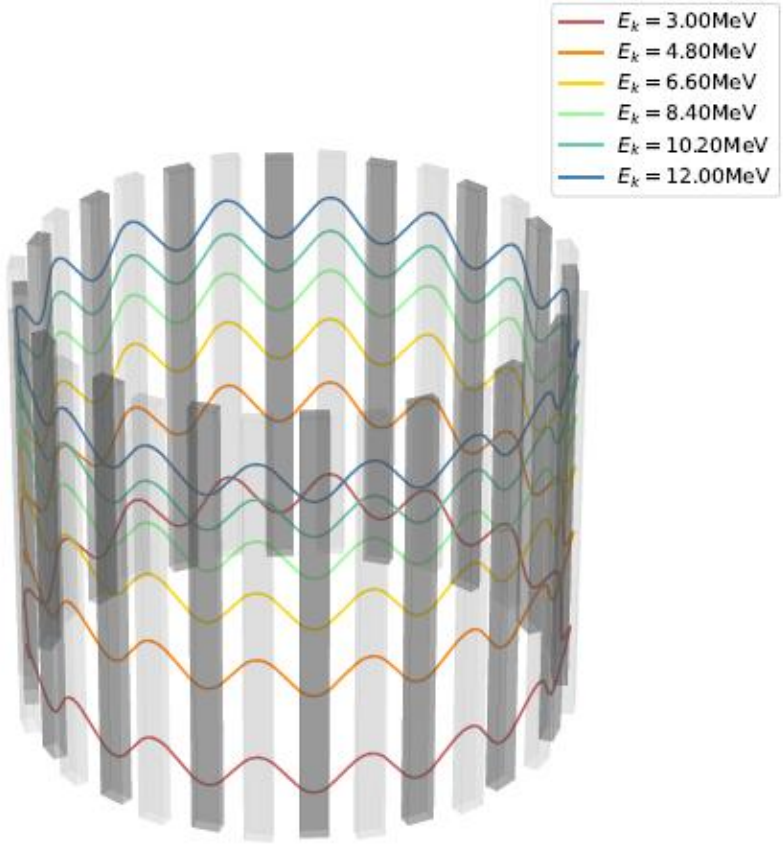



Figure 1.7: 3D view of an example scaling vFFA ring. Shaded volumes show the magnet geometry, with the darker grey shade representing the reverse bend magnets.

longitudinal dimension of the cavity.

In FFAs, due to the excursion of the beam during the acceleration cycle, it is necessary to construct cavities with wide transverse apertures and wide 'good field regions' - i.e. minimising transverse field components across the width of the region occupied by the beam during acceleration. Depending on the properties of the machine, and the energy ranges in which it operates, it may also be necessary to adjust the frequency of the accelerating field

# Recirculating Linear Accelerator (RLA) + Fixed Field Alternating Gradient (FFAG) = Fixed Field Recirculating Accelerator(FFRA)

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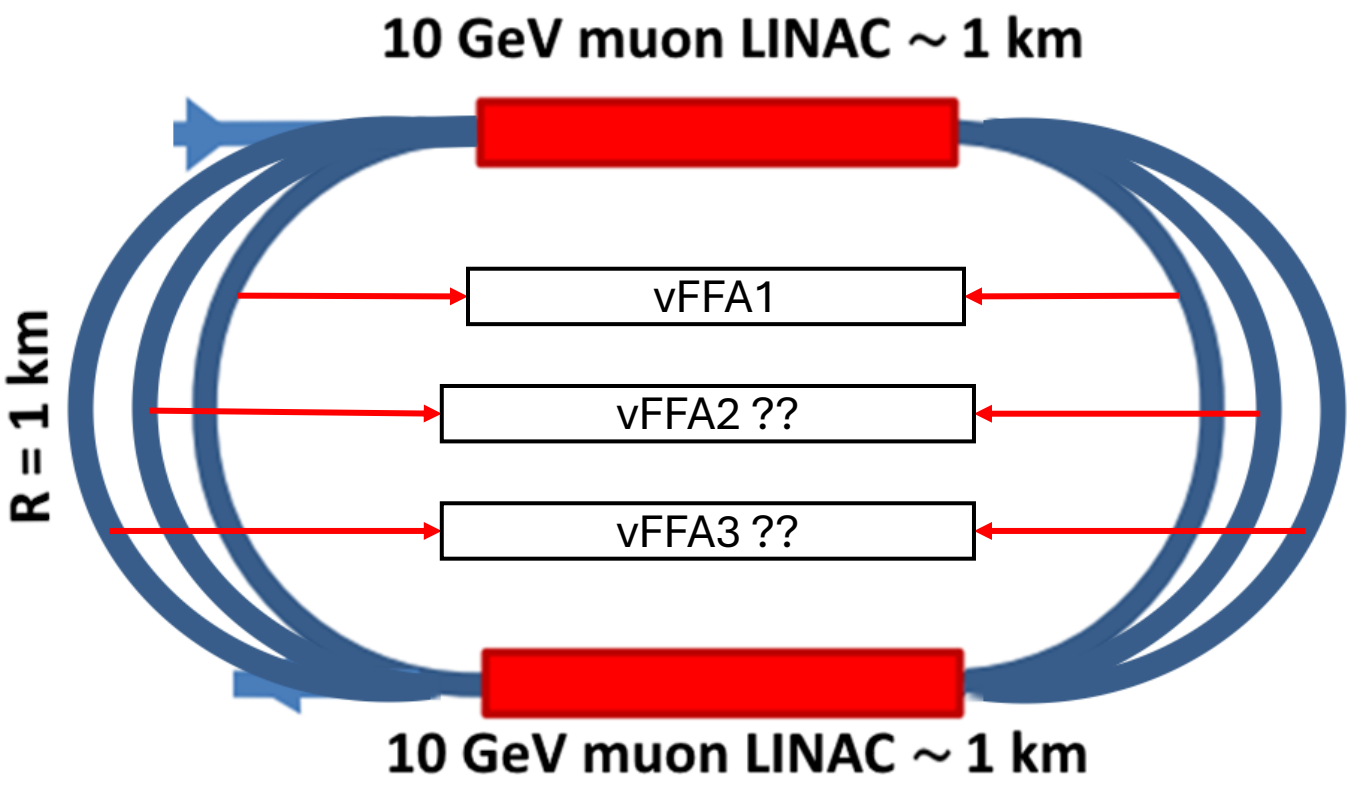


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325 GeV muons x 7 TeV protons

$\sqrt{s} = 3\ TeV$

Only vFFA1 for first stage

Ramp rate is 0 T/s !!

500 GeV muons can be possible with higher aperture size!!

# FIXED FIELD Tracking Code

We can try to simulate our racetrack shape in these codes after White Paper

## Appendix A

## The FIXFIELD Tracking Code

Throughout this work, numerical results are provided by simulations carried out using the FIXFIELD code [53]. FIXFIELD is a particle tracking code tailored specifically for simulation of scaling FFAs (though other types of magnets are also included).

The FIXFIELD tracking code, developed in C, is based on an implementation of a

chosen over SCODE as the primary simulation tool in this thesis due to the author's familiarity with C and relative lack of FORTRAN experience. Since 2019, implementations of vFFAs have now been added to numerous other codes, including OPAL [116] and Zgoubi [117].

The agreement of these simulations has since been cross-checked, and an example benchmarking of vFFA simulation results from FIXFIELD, SCODE, and OPAL can be found in [32] (Appendix A). Agreement is close across all three codes, differing by a max-

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# How many cold muons?

1/(20ms) where 20ms is the lifetime of the 1TeV muon

J-PARC like proton driver: 6.6 μC * 50 Hz * 2 bunches = 4.1 x $10^{15}$ protons/s

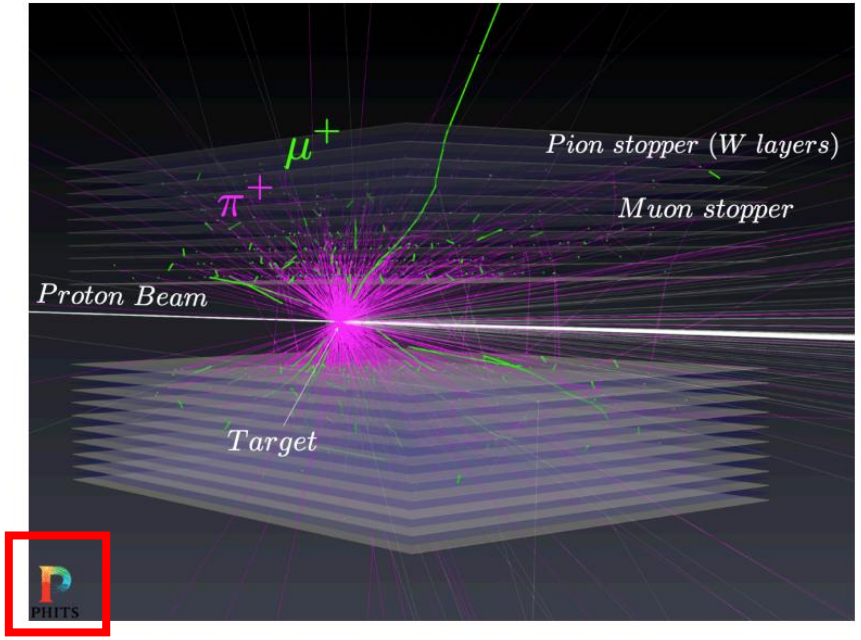
pion production target: 40 hits/bunch 0.016 $\pi^+$/proton 2.6 x $10^{15}$ $\pi^+$/s

pion stopping target: 0.5 stopping efficiency * 0.07 muons/$\pi^+$ 9 x $10^{13}$ $\mu^+$/s

$10^5$ larger than J-PARC MLF.

Super muon factory!

simulation: (Yoshida, Sakaki … in progress)




(Thermal muon production rate)

= (Muon stopping number on the layers)

× (MC correction for pion production)

× (Muonium formation)

× (Vacuum yield)

× (Loss of muoniums due to the decay)

PTEP 2022 (2022) 5, 053B02

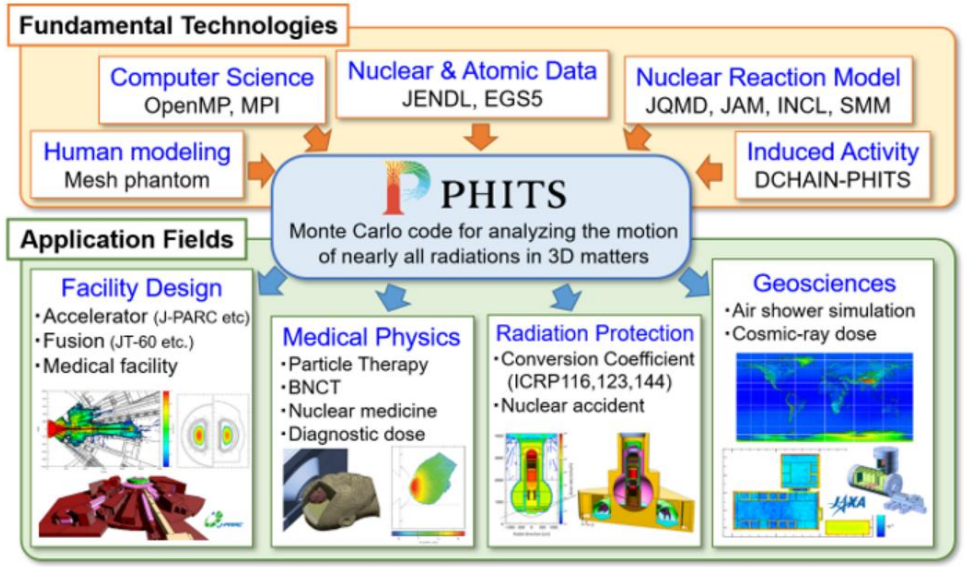
= 1.4 * $10^{-3}$ μ/p

→ **2.4 * $10^{14}$ μ/s** (J-PARC RCS: 6.6 μC, 2bunch 40 turns)

**O($10^{13}$ μ/s) will be available for collider experiment**

**What is PHITS?**

**PHITS** (**P**article and **H**eavy **I**on **T**ransport code **S**ystem) is a general purpose Monte Carlo particle transport simulation code developed under collaboration between JAEA, RIST, KEK and several other institutes. It can deal with the transport of all particles over wide energy ranges, using several nuclear reaction models and nuclear data libraries. PHITS can support your researches in the fields of accelerator technology, radiotherapy, space radiation, and in many other fields which are related to particle and heavy ion transport phenomena. (detail, pdf 5.3MB, sample_movie, pptx 35MB)




We can try to simulate muon production with PHITS and Geant4 after White Paper